

Day 7 — When the Trick Doesn't Help

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify which multiplications are well-suited to the *Ekādhikena/Anurūpyeṇa* family and which are not, and will articulate the conditions that make a different sūtra (*Nikhilam* or *Ūrdhva-tiryagbhyām*) more appropriate.

Materials

- Whiteboard.
- 8-problem comparison worksheet ($\times 11$ trick vs. long multiplication vs. *Nikhilam*-style).

Lesson flow

Warm-up (5 min)

- Quick chorus: 10 problems from Day 6's $\times 12$ – $\times 19$ family.

Core (20 min) — Where the trick breaks down

1. **Try $\times 23$.** “Let’s try the same proportional pattern with multiplier $23 = 20 + 3$.”
 - For 14×23 , would the trick scale?
 - $(14)(20 + 3) = 280 + 42 = 322$. Long way works.
 - The “proportional middle” pattern doesn’t extend cleanly past 19 because the multiplier has two non-trivial digits.
2. **Why?** Recall the identity: $\times(10 + k) \rightarrow 100a + 10(ak + b) + bk$. Clean because the “tens” part of the multiplier is **just 1**. For $\times 23$, the algebra is:

$$\begin{aligned} & (10a + b)(20 + 3) \\ &= 200a + 20b + 30a + 3b \\ &= 200a + 30a + 20b + 3b \\ &= (100 \cdot 2 + 100 \cdot 0 + \dots) - \text{no longer a tidy "outer-stays, inner-sums" pattern.} \end{aligned}$$

3. **The honest summary.**

The $\times 11$ trick (*Ekādhikena Pūrveṇa*) and its scaled cousins (*Anurūpyeṇa* for $\times 12$ to $\times 19$) work because the multiplier is *very close to 10*. Specifically, the multiplier must be $10 + k$ where k is a single digit.

For multipliers farther from 10 ($\times 23$, $\times 47$, $\times 98$), a DIFFERENT sūtra is the right tool — Tirthaji’s *Nikhilam* (“all from 9, last from 10”) for multipliers near 100, or *Ūrdhva-tiryagbhyām* (“vertically and crosswise”) for general multiplication.

4. **Connect to Module 9 / 11.** Future modules will cover *Nikhilam* and *Ūrdhva-tiryagbhyām* in depth. Today we just note their existence.

5. **The bigger lesson.** “Every mental-math shortcut has a domain. Knowing *WHEN* a trick applies is part of being a fluent mathematician. The standard long-multiplication algorithm always works; the shortcuts are faster only inside their domain.”

Activity (15 min) — Comparison drill

- 8 problems. For each, decide *before* solving: would you use (A) the $\times 11$ family trick, (B) long multiplication, or (C) a different mental shortcut you can articulate?

- | | |
|-------------------|--------------------|
| 1. 67×11 | 5. 84×23 |
| 2. 47×13 | 6. 99×99 |
| 3. 56×19 | 7. 25×12 |
| 4. 38×47 | 8. 102×11 |

- Pair work. Solve and discuss your method choice.
- Answers and “best method” given on the back.

Wrap-up (5 min)

- Class compiles “the $\times 11$ family domain”: all 2-digit numbers $\times (11, 12, 13, \dots, 19)$ — about $90 \times 9 = 810$ distinct problems. That’s a significant mental-math territory.
- Preview Day 8: “Tomorrow — where do these 16 sūtras come from? The honest history.”

Student-facing content

The domain of the $\times 11$ trick

The “split and add” trick works for multipliers of the form $10 + k$ where k is a single digit (so $\times 11$ through $\times 19$).

For other multipliers, use: - Long multiplication (always works). - *Nikhilam* “all from 9, last from 10” — when both numbers are near 100 (e.g. 98×97). - *Ūrdhva-tiryagbhyām* “vertically and crosswise” — general multiplication.

Knowing **when to use** a trick is as important as knowing **how**.

Homework

- Find any 3 multiplication problems in your homework or a textbook. For each, decide which method ($\times 11$ family / long / other) you’d use and **why**. Write 1 sentence of justification per problem.

Differentiation

- Tier up:** Find any 2-digit $\times$ 2-digit multiplication where the $\times 11$ trick is NOT optimal but a different mental-math shortcut would beat long multiplication. (Examples: $25 \times 24 = 25 \times 24 = 600$ via $25 \times 4 \times 6$; $75 \times 12 = 900$ via $75 \times 12 = 75 \times 12$.)
- Tier down:** Focus on identifying $\times 11$ vs. non- $\times 11$ cases. Skip the “best alternative method” question.

Teacher notes

- This day is short by design — it's the honesty hour. Don't over-fill.
- Some students will resist the framing "*the trick has a limited domain*" and try to force-fit ×23 into a pattern. Honour the impulse, but redirect: "*You'd be inventing your own technique — which is great, but it's not what this trick does.*"
- Mention that Tirthaji's 16 sūtras collectively cover a wide territory; ×11 is just one of them. This sets up Day 8.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra list (especially *Nikhilam* — sūtra 2, and *Ūrdhva-tiryagbhyām* — sūtra 3).
- *Microsoft Word - sutras1.pdf* — full sūtra list.